Solutions to Quiz 1

1 - Quiz

In a linear regression, the dependent variable is always on the vertical axis.

30 sec

- True ✓
- False ✗

Dependent : y

Independent : x

2 - Quiz

Which of the following correlation coefficients indicates the strongest linear relationship?

30 sec

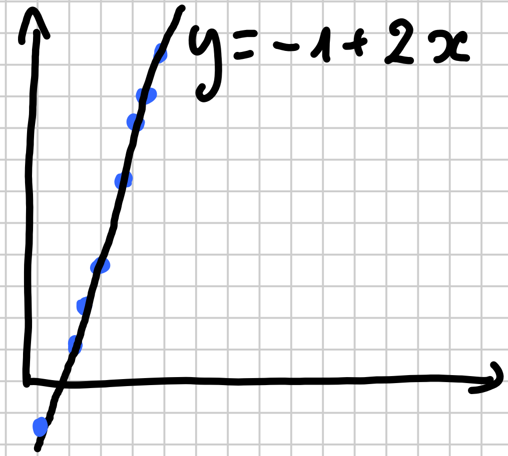
-0.65	✗
-0.77	✓
0	✗
-0.12	✗

3 - Quiz

If all the points in a scatterplot are aligned on the line y=-1+2x, then the correlation r is

60 sec

- 0 ✗
- 1 ✓
- -1 ✗
- It depends. ✗



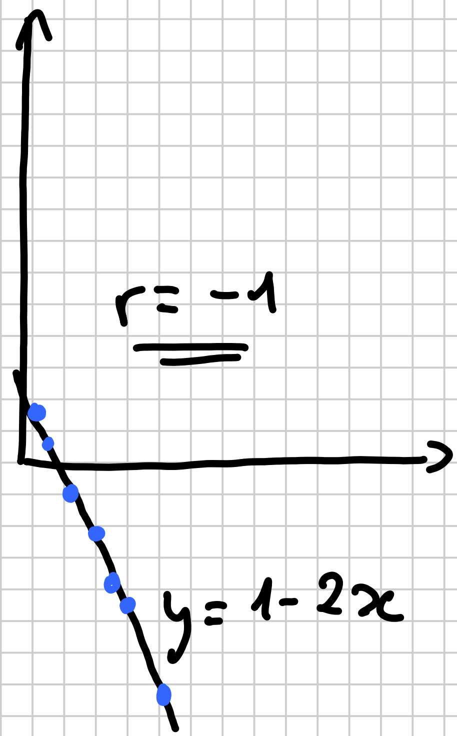
$\underline{r = 1}$

4 - Quiz

If all the points in a scatterplot are aligned on the line y=1-2x, then the correlation r is

60 sec

- 0 ✗
- 1 ✗
- -1 ✓
- It depends. ✗



5 - Quiz

The regression equation is $\hat{y} = 35.2 + 5.3\,x$. If x=100 and y=600, then

90 sec

- the predicted value is 600 and the residual is 0; ✗
- the predicted value is 565.2 and the residual is 0; ✗
- the predicted value is 565.2 and the residual is 34.8; ✓
- neither of the previous answers is correct. ✗

$$\hat{y} = 35.2 + 5.3 \times 100$$
$$= \underline{\underline{565.2}}$$
$$\hat{\varepsilon} = y - \hat{y} = 600 - 565.2$$
$$= \underline{\underline{34.8}}$$

6 - Quiz

The error terms in a simple linear regression must be

60 sec

- normally distributed with mean 0 and variance 1;
- normally distributed with mean 0 and constant variance;
- normally distributed with constant mean and variance 1;
- neither of the previous answers is correct.

7 - Quiz

For a given dataset S_{xy}=2 and S_{xx}=1. What is the slope in the linear regression equation?

30 sec

- 0.5 ✗
- 2 ✓
- insufficient information provided to answer the question. ✗

Cov(x,y)

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} = \frac{2}{1} = \underline{\underline{2}}$$

Var(x)

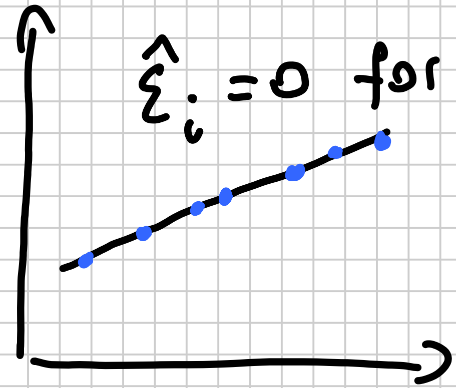
8 - Quiz

If all the points are aligned on the regression line, the residuals are all equal to

60 sec

- 0 ✓
- 1 ✗
- insufficient information provided to answer the question. ✗

$\hat{\varepsilon}_i = 0$ for all i



9 - Quiz

The regression equation is $\hat{y} = 0.2 + 0.3\,x$. The correlation between x and y is:

60 sec

- 0.3 ✗
- -0.3 ✗
- I do not know and am very scared. ✓

$$r = \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}} = ?$$

(we only know

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}}\Big)$$

10 - Quiz

For a given observation, y_i=0.5 and the residual is 0.5. The predicted value $\hat{y}_i$ is

60 sec

- 1
- 0
- neither of the previous answers is correct.

$$\hat{\varepsilon}_i = y_i - \hat{y}_i$$

$$= 0.5 - \hat{y}_i = 0.5$$

$$\rightarrow \underline{\underline{\hat{y}_i}} = 0$$

11 - Quiz

If the correlation coefficient is r=0, then

60 sec

- there is no relation between x and y; ✗
- there is no linear relation between x and y; ✓
- there is no nonlinear relation between x and y. ✗

Could be :